

DATE : 21.09.2026
DT/NT : DT
LESSON : LINUX
SUBJECT: LINUX ESSENTIALS
SESSION : 1
BATCH : B 424-020DE

AWS-DEVOPS



TECHPRO
EDUCATION



techproeducation.com



+1 (585) 304 29 59



Content reviewed: 5/9/2026

Introduction to Linux

- Linux Essentials for AWS & DevOps



What We Will Learn



- ▶ Explain why Linux matters in cloud and DevOps environments.
- ▶ Distinguish the Linux kernel from a Linux distribution.
- ▶ Recognize current distributions used in AWS and enterprise environments.
- ▶ Explain terminal, shell, command prompt, and command execution.
- ▶ Navigate the filesystem and use essential shell commands.
- ▶ Use filename globbing and brace expansion correctly.
- ▶ Apply the concepts in a short AWS EC2 Linux lab.

Why Linux?

Where Linux?

Why DevOps?



Why Linux?



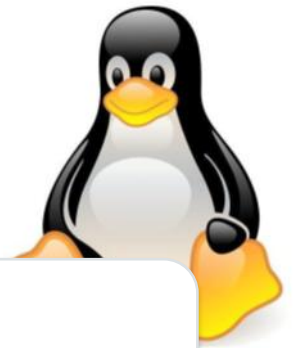
Technical reasons

- ▶ Stable server platform
- ▶ Strong permissions and security model
- ▶ Powerful CLI and scripting
- ▶ Mature networking and package tools

DevOps reasons

- ▶ Cloud and container ecosystem
- ▶ Automation-friendly
- ▶ Remote administration over SSH
- ▶ Broad open-source tooling and community

Where Linux Runs Today



Cloud VMs

EC2 and other cloud virtual machines

Containers

Docker / containerd workloads

Kubernetes

Control-plane and worker nodes

Web & APIs

NGINX, Apache, application servers

Data Services

Databases, caches, message brokers

Automation

CI/CD runners and build agents

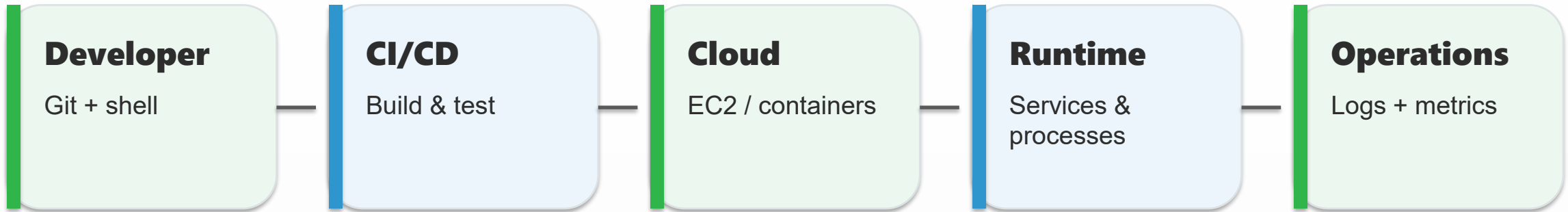
Networking

Routers, firewalls, appliances

Devices

Android kernel, IoT and embedded systems

Linux in an AWS / DevOps Workflow



Linux is the common operational layer connecting development, automation, cloud infrastructure, and production troubleshooting.

Linux Evolution

.



Linux Kernel vs. Linux Distribution



Linux kernel

The core that manages CPU, memory, devices, processes, networking, and system calls.

Linux distribution

A complete operating system built around the Linux kernel plus utilities, libraries, package management, and applications.

In everyday conversation, people often say “Linux” to mean a complete Linux-based operating system.

Android uses the Linux kernel, but it is not a typical GNU/Linux server distribution.

Linux System Stack



Applications & Services

Shell / CLI

Utilities & Libraries

Linux Kernel

Hardware

The shell is not the kernel. It is one user-space interface for interacting with the operating system.

Linux Timeline — Key Milestones



UNIX begins

GNU
development

Linux + GPL
v2

Linux 1.0

WSL 2

1969

1984

1991

1994

2019

1983

GNU
announced

1989

GPL v1

1992

Linux under
GPLv2

2016

WSL

2023

Amazon Linux
2023

GNU was announced in 1983 and development began in 1984; WSL appeared in 2016 and WSL 2 in 2019.

Linux Distributions for AWS / DevOps



Family	Examples	Package tools	Why it matters
Debian family	Debian, Ubuntu	apt / dpkg	Cloud servers, development
Red Hat family	RHEL, Fedora, CentOS Stream	dnf / rpm	Enterprise Linux
RHEL-compatible	Rocky Linux, AlmaLinux	dnf / rpm	Enterprise-compatible servers
AWS	Amazon Linux 2023	dnf / rpm	AWS EC2 workloads
Other	openSUSE	zypper / rpm	General Linux knowledge

CentOS Linux 7 is end-of-life; the CentOS project now focuses on CentOS Stream.

Open Source & the Linux Ecosystem



What Is Open Source?



Open-source software makes its source code available under a license that permits use, inspection, modification, and redistribution according to that license.

Source available

You can inspect how the software works.

Modify & improve

Licenses define how changes can be made and shared.

Collaborate

Communities and organizations can build together.

Open source does not simply mean “free of cost.” License terms and user freedoms are the important part.

FSF and OSI — Related, but Not Identical



Free Software Foundation (FSF)

Emphasizes software freedom and users' rights to run, study, modify, and share software.

Open Source Initiative (OSI)

Maintains the Open Source Definition and approves licenses that meet its criteria.

Shared ground

- ▶ Source code can be inspected.
- ▶ Modification and redistribution are allowed under license terms.
- ▶ Open collaboration is a major part of the ecosystem.

Package Management — Current View



Debian / Ubuntu

```
sudo apt update  
sudo apt install git  
sudo apt remove git
```

High-level tool: apt

Low-level package format/tooling: .deb / dpkg

RHEL / Fedora / Amazon Linux 2023

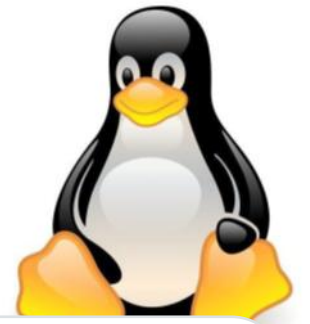
```
sudo dnf check-update  
sudo dnf install git  
sudo dnf remove git
```

High-level tool: dnf

Low-level package format/tooling: .rpm / rpm

Amazon Linux 2023 uses DNF by default. The yum command remains available as a compatibility pointer to dnf.

Open-Source Tools Commonly Seen in DevOps



Git

Version control

Ansible

Configuration automation

Kubernetes

Container orchestration

NGINX

Web / reverse proxy

Prometheus

Metrics

Grafana

Visualization

PostgreSQL

Database

OpenTofu

Infrastructure as code

The point is not to memorize tools today—notice how much of the modern DevOps ecosystem is designed to run naturally on Linux.

Using Linux on Different Platforms



AWS EC2 + Amazon Linux 2023



- ▶ A Linux virtual machine running in AWS.
- ▶ A practical environment for learning real server administration.
- ▶ Amazon Linux 2023 is an AWS-maintained Linux distribution.
- ▶ AL2023 uses dnf as its default package manager.

Useful first checks

```
whoami  
cat /etc/os-release  
uname -r  
dnf --version
```

Use EC2 when you want to practice SSH, services, networking, permissions, logs, and deployments on an actual cloud VM.

Windows Subsystem for Linux (WSL)



What it is

A Linux environment integrated into Windows, useful for local command-line and DevOps practice.

When to use it

Local shell practice, Git, scripting, CLI tools, and Linux workflows without a separate VM.

```
# Run in PowerShell  
wsl --install
```

Linux in a Virtual Machine



- ▶ Runs a complete guest operating system on your computer.
- ▶ Works with tools such as VirtualBox, VMware, or Hyper-V.
- ▶ Useful for snapshots, isolated labs, and experimenting safely.
- ▶ Uses more local CPU, memory, and disk than a typical WSL setup.

Best for

Full OS behavior, isolated networking, multi-node labs, snapshots.

AWS CloudShell



Browser-based shell

Launch a shell directly from the AWS Management Console.

AWS-ready

AWS CLI is available and the shell uses credentials from your console session.

Know the limit

Great for AWS CLI tasks; not a substitute for learning full EC2 server administration.

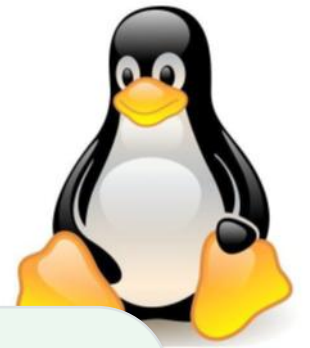
```
aws sts get-caller-identity
pwd
ls -la
```

Shell & Terminal

Understand the interface before memorizing commands.



Terminal vs. Shell



Terminal

The interface/window where you type commands and see output.

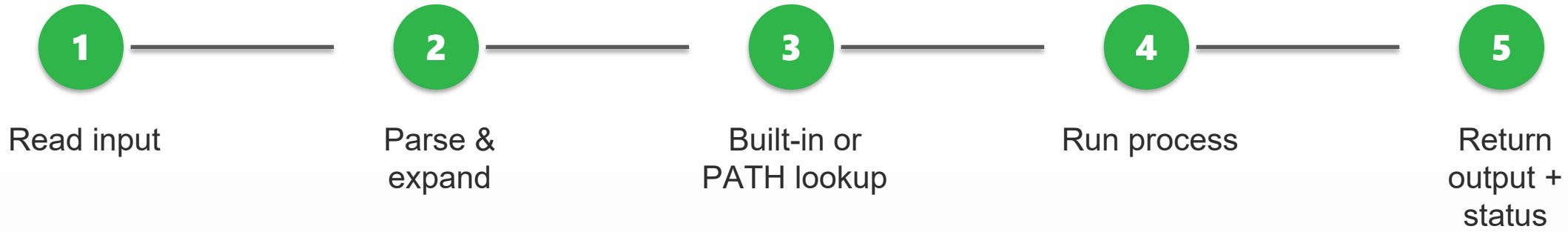
Shell

A command interpreter that parses commands, performs expansions, and runs built-ins or external programs.

Common shells: `bash` • `sh` • `zsh`

Bash is widely used, but Linux does not have one universal “standard shell.”

How the Shell Executes a Command



Useful discovery commands

```
type cd
type ls
command -v python3
```

Command Prompt Anatomy



student

web-01

~/project

\$

```
student@web-01:~/project$
```

current user

host name

current directory

normal user

\$ → normal/unprivileged shell | # → commonly indicates a root shell

Do not rely only on the prompt symbol for security decisions—verify identity with `whoami` or `id`.

Linux Filesystem & Basic Shell Commands

Navigation first; then files, directories, and patterns.



Filesystem Navigation Basics



Important path symbols

- ▶ / → filesystem root
- ▶ ~ → current user's home
- ▶ . → current directory
- ▶ .. → parent directory

Absolute vs. relative paths

Absolute path

Starts from /
Example: /var/log

Relative path

Starts from where you are now
Example: logs/app.log

Case Sensitivity & Hidden Files



Case matters

App.log, app.log, and APP.log are different names on common Linux filesystems.

Hidden by convention

Names beginning with a dot are normally hidden from a plain ls listing.

```
touch App.log app.log
touch .env
ls
ls -a
```

Common hidden files/directories: .bashrc .ssh .gitconfig .env

General Command Structure



command

argument

```
ls -lah /var/log
```

options

- ▶ Options change command behavior.
- ▶ Arguments tell the command what to work on.
- ▶ Use command --help for quick usage help.
- ▶ Use man command for the manual page when available.

Navigation Commands — pwd and cd



```
pwd  
cd /  
cd ~  
cd ..  
cd -
```

- ▶ pwd → print current working directory
- ▶ cd / → go to filesystem root
- ▶ cd ~ or cd → go to your home directory
- ▶ cd .. → move one level up
- ▶ cd - → switch to the previous directory

Practice reading the prompt and checking pwd frequently until path navigation becomes automatic.

Listing & Creating Directories



ls — list directory contents

```
ls
ls -l
ls -a
ls -lah
```

- -l details
- -a hidden entries
- -h human-readable sizes

mkdir — create directories

```
mkdir project
mkdir -p app/config
mkdir -v logs
```

- -p creates parent directories as needed
- -v shows what was created

touch — More Than “Create a File”

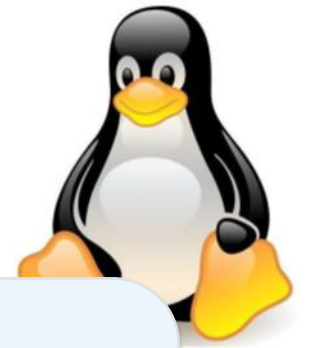


By default, touch updates a file's access and modification timestamps. If the file does not exist, it normally creates an empty file.

```
touch app.log
ls -l app.log

# Run touch again later
touch app.log
stat app.log
```

Globbing vs. Brace Expansion



Globbing (filename expansion)

The shell matches existing pathnames using patterns such as *, ?, and [].

Globbing

```
ls *.log  
ls file?.txt  
ls [a-c]*
```

Brace expansion

The shell generates strings from a pattern such as {1..5} or {dev,test,prod}. It does not require files to already exist.

Brace expansion

```
touch file{1..5}.txt  
mkdir -p {dev,test,prod}/logs
```

The shell performs these expansions before the command itself receives the resulting arguments.

AWS EC2 Mini-Lab — Day 1



- ▶ Connect to a Linux EC2 instance.
- ▶ Identify the current user, host, distribution, and kernel.
- ▶ Create a workspace and navigate through it.
- ▶ Create a file and a nested directory.
- ▶ List normal and hidden entries.

```
whoami
hostname
pwd
cat /etc/os-release
uname -r
ls -la
mkdir -p linux-lab/config
cd linux-lab
touch app.log
ls -lah
```

Day 1 Challenge



Without copying the commands from the previous slide, complete these tasks:

- ▶ Create `~/day1/{logs,config,backup}` using one command.
- ▶ Create files `app1.log` through `app5.log` using brace expansion.
- ▶ List only the `.log` files using a glob pattern.
- ▶ Create a hidden file named `.env` and prove it is hidden from plain `ls`.
- ▶ Move to `/var/log` using an absolute path, then return to the previous directory with `cd -`.
- ▶ Show your Linux distribution and kernel version.

Done condition: you can explain what each command does—not just reproduce it.

Day 1 Key Takeaway

Linux is the operational foundation behind much of cloud and DevOps.

Today you learned how the system is organized and how to start navigating it from the shell.

Do you have any questions?

